

Predicting Airline Passenger Satisfaction



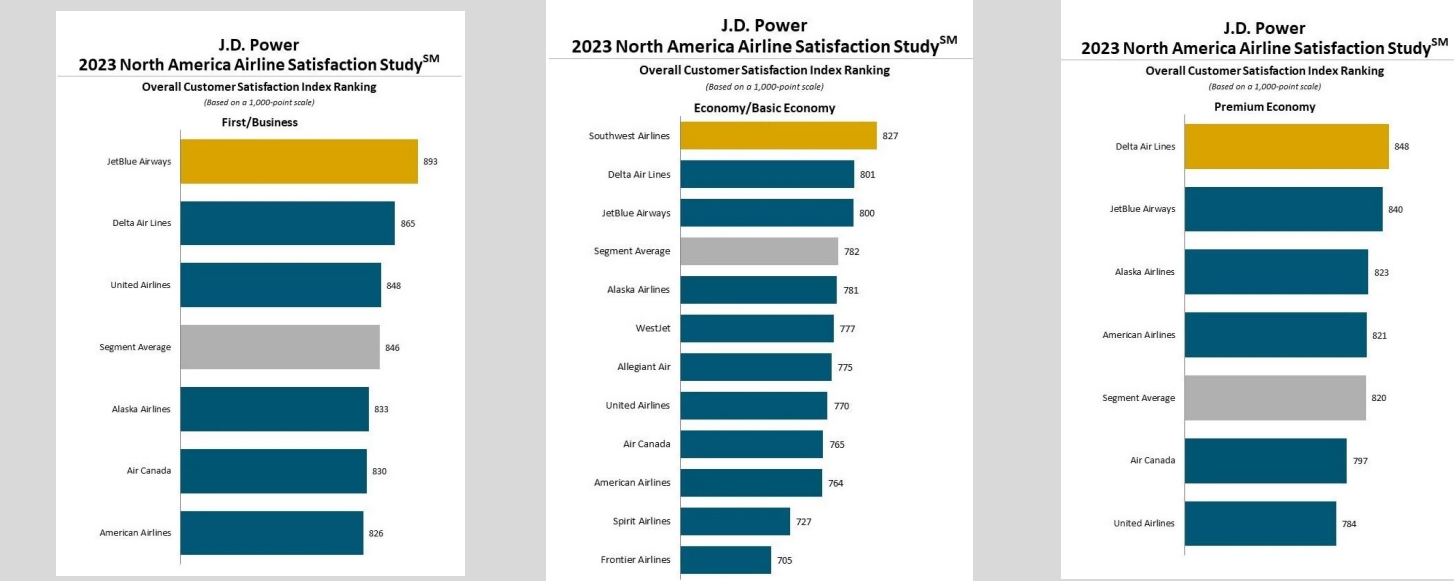
Eric Tysinger
Purdue University, Daniels School of Business
etysinge@purdue.edu

ABSTRACT

I built a predictive model that analyzes numerous variables such as type of travel, inflight entertainment, seat comfort, online boarding, departure delay in minutes, and many more, to predict whether or not the passenger was satisfied with their traveling experience. Commercial airlines are constantly looking to gain a competitive advantage in such a competitive industry and one way to do that is provide the ideal traveling experience. By understanding the significant factors that yield a great traveling experience, I hope airlines can use this model to more effectively and efficiently devise their money and resources to improve their passenger experience.

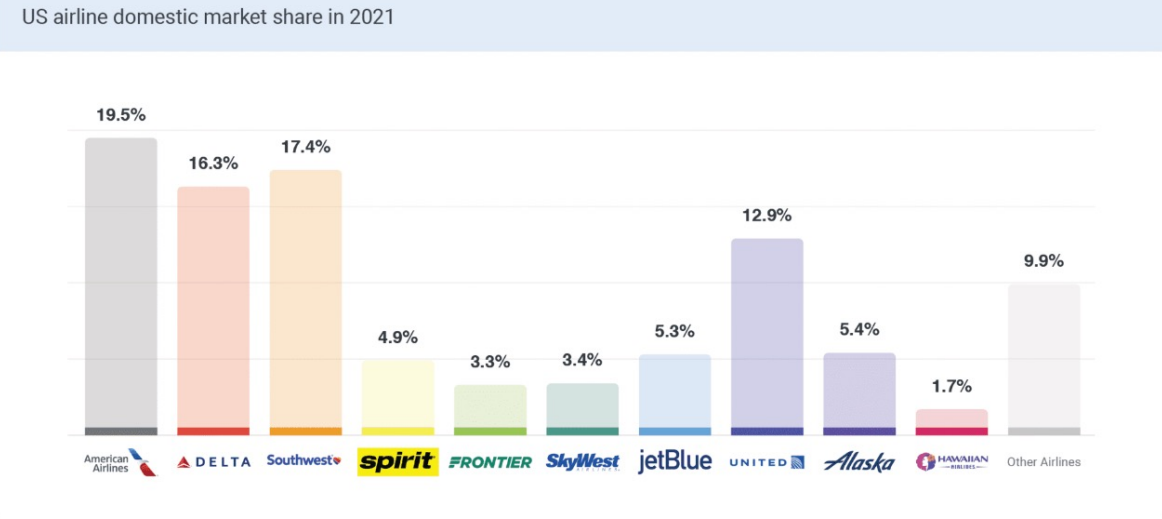
BUSINESS PROBLEM

My goal is to help airlines gain a competitive advantage in the industry by efficiently and effectively allocating their money and resources to provide the best traveling experience possible. According to a 2023 study by JD Power, passenger satisfaction has decreased in 2023, mostly due to the cost of the airfare. So, if airlines truly understand what the main drivers are of passenger satisfaction, then they can use their resources efficiently to improve the passenger experience. In addition, some airlines thrive when it comes to economy class, and some more in first/business class, as illustrated by the graphs below. So, each airline may need to focus their time and resources on a certain flight class and understand their strengths and weaknesses.



<https://www.jdpower.com/business/press-releases/2023-north-america-airline-satisfaction-study>

The bottom line is the commercial airline industry is an extremely competitive market, as per the figure from the NYC Data Science Academy below, so a competitive advantage in quality of traveling experience could go a long way. My project will be successful if



<https://nycdatascience.com/blog/student-works/us-airline-industry/>

we can predict with 90% accuracy whether or not a passenger is satisfied with their traveling experience, based on numerous variables that we have collected



ANALYTICS PROBLEM FRAMING

- Our measured target variable is airline passenger satisfaction. We want to predict passenger satisfaction. A classification predictive model would allow us to predict whether or not a passenger was satisfied with their traveling experience. Since we want to reduce passenger dissatisfaction and increase passenger satisfaction, given our predictive model, we can infer the variables that are contributing the most to passenger dissatisfaction, and reduce those as much as possible, and we can infer the variables that are contributing most to passenger satisfaction, and increase those as much as possible.
- We are seeking the model that most accurately predicts customer satisfaction, as that will give as the best idea of which predictors contribute and do not contribute significantly, to passenger satisfaction.

RESEARCH QUESTIONS

- What are the significant predictors of airline passenger satisfaction?

DATA

Data Collection/Overview:

Dataset is from Kaggle. It has 26,000 values, and 24 variables:

- 14 of the variables are responses from a survey with a class of 1-5; 1 being bad, and 5 being great. I changed those variables from continuous to categorical since I felt they served more as categories on a number line rather than an actual numeric value.
- The survey response variables include inflight wifi service, departure/arrival time convenience, ease of online booking, gate location, food and drink, online boarding, seat comfort, inflight entertainment, on-flight service, leg room service, baggage handling, checkin service, inflight service, and cleanliness
- There were a few other categorical variables including gender, customer type, type of travel, and class.
- Then there were a few continuous variables such as age, flight distance, departure delay in minutes, and arrival delay in minutes

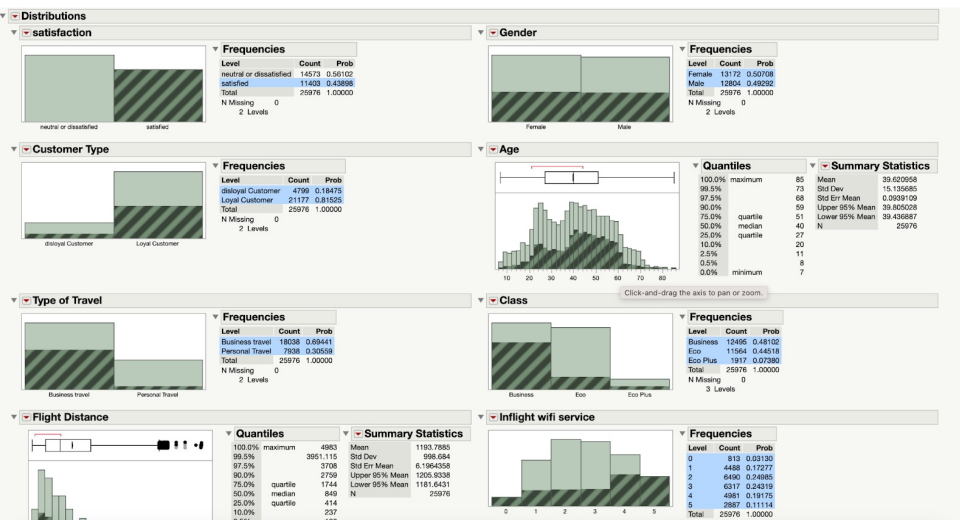
Data preprocessing clear

Missing values and outliers:

- I changed all the variables that had responses from 1-5 from a survey, to categorical variables rather than continuous.
- There were no outliers per JMP pro
- There were 83 missing values in the “Arrival Delay in Minutes” column, so I imputed missing values for that column. Then I standardized the four numeric variables.

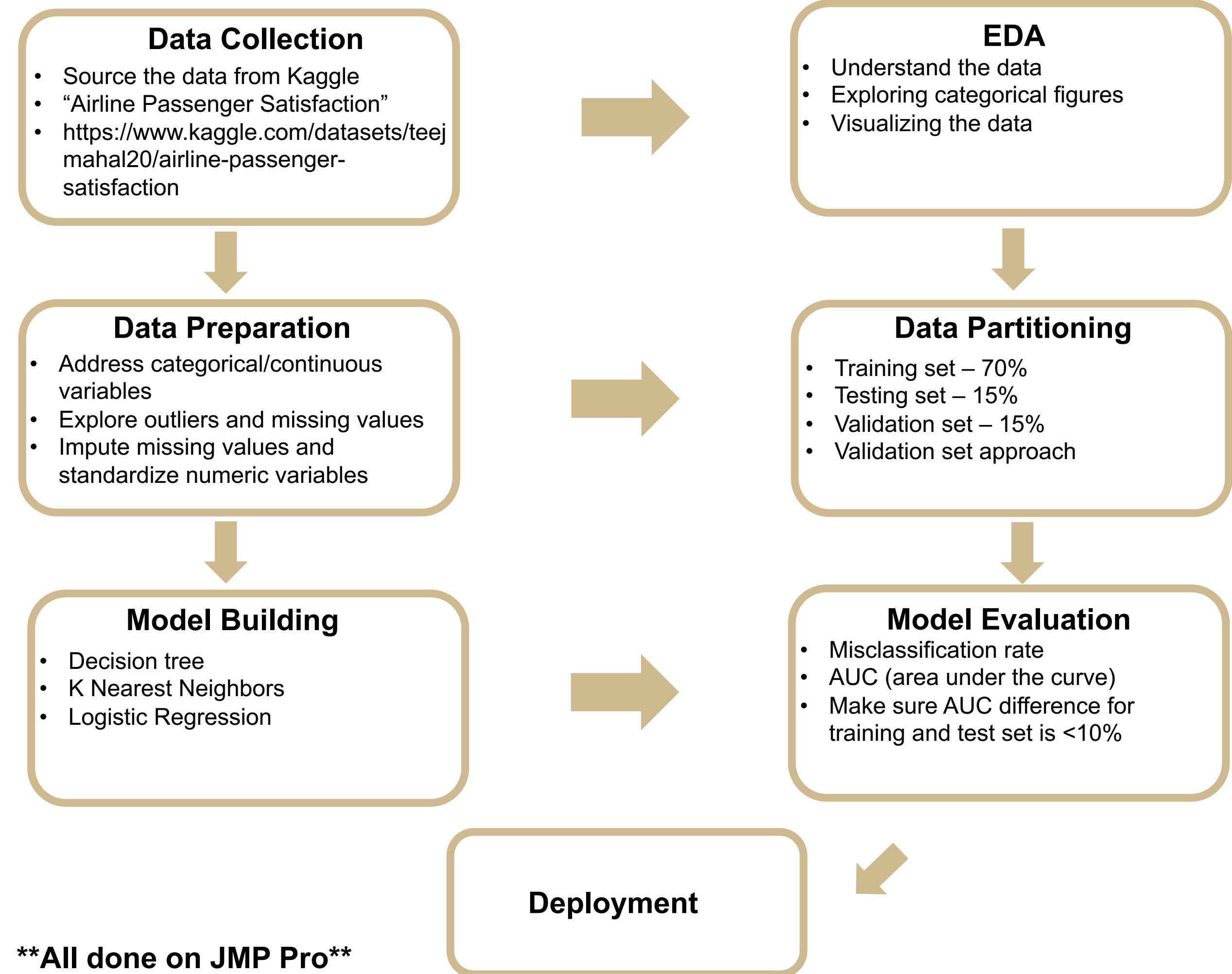
Notable Data Relationships:

- Here, I created distributions for all the variables, and I highlighted satisfaction just to get a gauge of strong predictors of passenger satisfaction. From this, I noticed that class, type of travel, online boarding, and seat comfort, were all significant predictors of passenger satisfaction.



Mitchell E. Daniels, Jr.
School of Business

METHODOLOGY

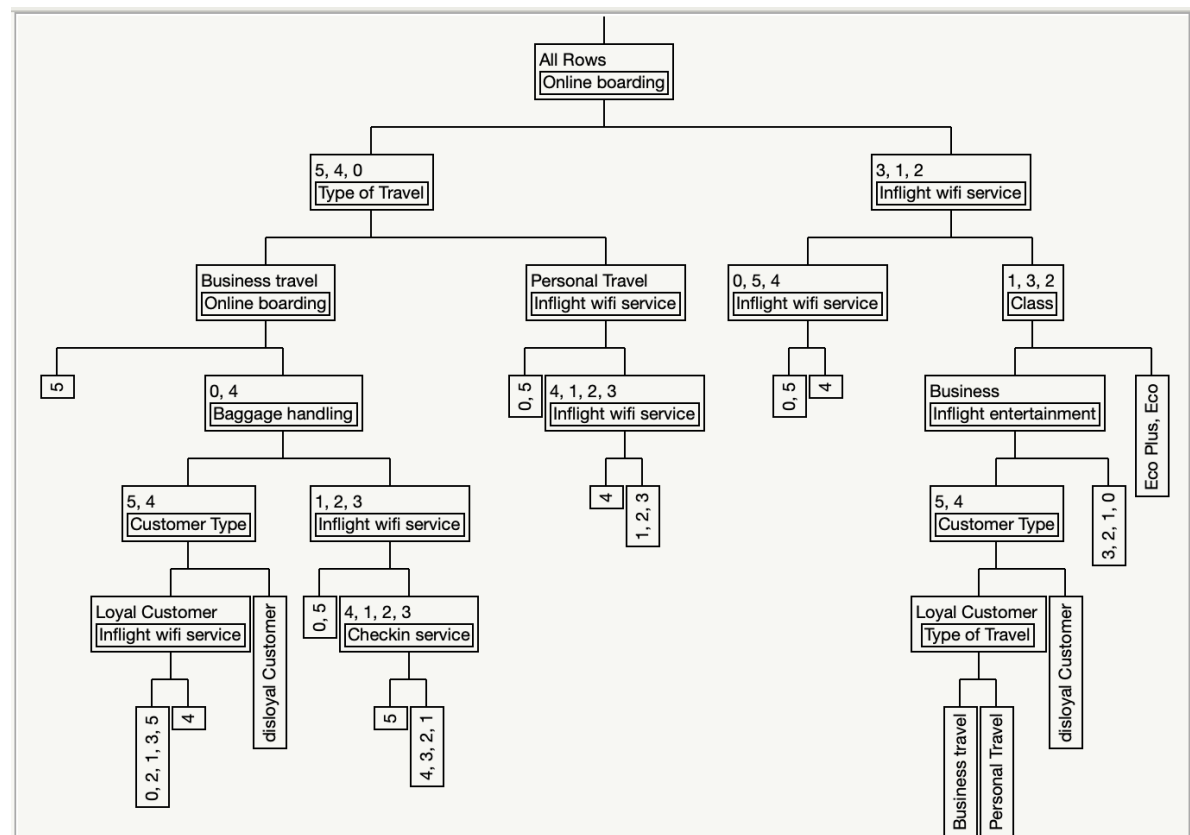


****All done on JMP Pro****

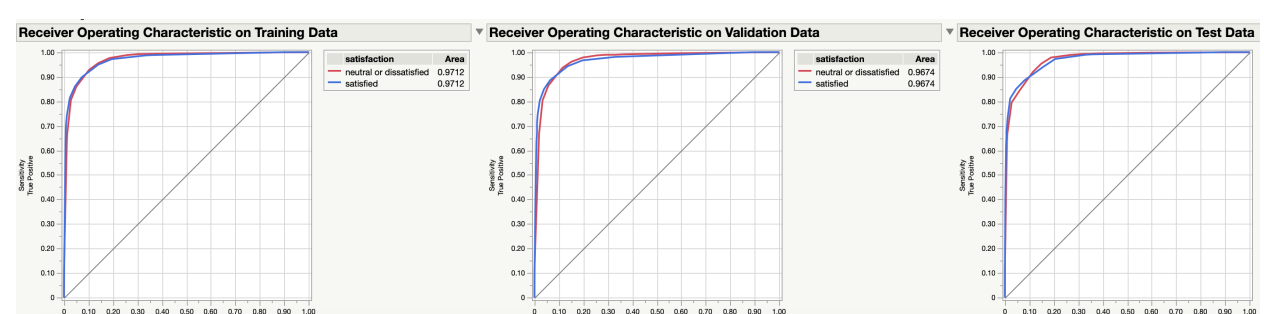
MODEL BUILDING AND EVALUATION – STATISTICAL PERFORMANCE

Decision Tree:

- At first, the Rsquare value was high, but the training and test R-square were too far apart (>0.1) indicating that the dataset was overfit, so I slowly pruned the data until they were close enough. The ROC Curves all hover around 0.97, indicating a well fit model. In addition, the misclassification of all sets are less than 10%, which meets our 90% accuracy goal was met.



	RSquare	N	Number of Splits
Training	0.701	20204	16
Validation	0.684	2886	
Test	0.693	2886	



Misclassification Rate 0.0854 0.0859 0.0953

Looking at the decision tree, you can see that online boarding rating was the most significant predictor, followed by inflight wifi service, and the type of travel. Airlines should focus their time and resources on these factors. Some notable omitted variables that can be pushed to the side include food and drink, baggage handling, and seat comfort.

Nominal Logistic Fit:

Source	LogWorth	PValue
Type of Travel	270.883	0.00000
Customer Type	199.214	0.00000
Inflight wifi service	165.435	0.00000
Online boarding	147.435	0.00000
Checkin service	45.632	0.00000
Seat comfort	43.144	0.00000
Departure/Arrival time convenient	29.138	0.00000
Inflight service	24.346	0.00000
Leg room service	24.284	0.00000
Baggage handling	20.710	0.00000
On-board service	19.808	0.00000
Ease of Online booking	16.945	0.00000
Inflight entertainment	15.318	0.00000
Class	12.762	0.00000
Cleanliness	9.455	0.00000
Gate location	6.463	0.00000
Food and drink	2.387	0.00401
Std Imputed_Arrival Delay in Minutes	1.619	0.02406
Gender	1.539	0.02890
Std Age	0.361	0.43510
Std Flight Distance	0.191	0.64434
Std Departure Delay in Minutes	0.085	0.82264

Entropy RSquare	Misclassification Rate	AUC
0.7266	0.0693	0.9776

The Rsquare values are consistent between the training and testing sets, both between 0.72-0.74. In addition, the AUC values are also very close to 1, and off by less than 0.01. Finally, the misclassification rate is well below 10%.

MODEL EVALUATION – BUSINESS IMPLICATIONS

Given the nominal logistic regression and decision tree models, commercial airlines can understand the most significant predictors of passenger satisfaction. With this information, they can allocate their resources, and money to improve upon the important factors, to significantly improve their passenger satisfaction. This increased passenger satisfaction will give them a competitive advantage in the industry as passengers will be more willing to pay the higher fares as discussed in earlier, which will lead to increased sales in future quarters. As for how much those sales is yet to be determined, but with such a competitive market, any advantage is a good one.

CONCLUSIONS

With these models, commercial airlines can better understand their passengers wants and needs, so that they can provide the best traveling experience possible.

- The Decision Tree and Nominal Logistic Model predicted airline passenger satisfaction with >90% accuracy
- The greatest predictors of passenger satisfaction are the type of travel, inflight wifi service, and online boarding. Flight distance, age of the passenger, gender, and flight delay are insignificant.

It may be worth removing the continuous predictors from the data altogether, as they posed no significance to the model.